

Higher Reflection Identities and Resonant Zeta Calculus

Complex-power sine kernels, harmonic parity,
geometric collisions, and quartic Tornheim derivatives

Research continuation for the ProveIt project

11 October 2026 (UTC)

Abstract

We develop four exact continuations of the current ProveIt polylogarithm manuscript and its newest incoming reports. A complex-power sine weight produces a two-parameter Hurwitz transform with explicit Gamma Fourier coefficients and a polylogarithmic Euler integral. Removing its complete crossing polar term gives every finite-part moment of an argument derivative of a generalized Stieltjes function against a logarithmic sine power. The first nonlinear specialization identifies a combination of the digamma cube and a logarithmic sine-square moment in ordinary zeta and Stieltjes constants. A second result characterizes exactly the polynomials in generalized harmonic numbers whose centered Dirichlet series have no poles at any nonpositive even integer. A third gives an explicit analytic subtraction for arbitrary finite clusters of periodic digamma factors, including hierarchical collisions. Finally, we compute the complete fourth directional derivative of Tornheim zeta, give convergent representatives for its three formal coordinates, and prove an all-order dimension theorem for the specified symmetry and boundary relations. It explains why one diagonal coordinate suffices for cyclic cubic and quartic sums and why the same argument fails at order five. All statements include their continuation or finite-part conventions. The package contains reproducible exact algebra and independent numerical diagnostics. The Gaussian S_6 and revised S_8 evaluations remain open.

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1 Scope, provenance, and conventions

The starting point is the ProveIt repository at revision

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The source inventory under `Analysis/Polylogarithms/docs/manuscript` and all nine ZIP reports present under `docs/incoming` at that revision were retrieved. Selected canonical manuscript sections were read alongside the reports. The focused reading concerned the integration, harmonic, reflection, Stieltjes contact, and Tornheim sections. This is a continuation of those parts, not a claim to have audited every proof in the four-volume project. The accompanying `SOURCE_AUDIT.md` identifies the relevant files and the exact boundary between inherited results and the present extensions.

The latest incoming reports matter mathematically. They already prove the cubic digamma–harmonic bridge, the independent cubic Tornheim formula, the polynomial reflected Stieltjes closure, and several all-depth spectral completion results. Repeating any of these as a new theorem would misrepresent the state of the project. The questions continued here are more specific:

1. The nonpolynomial reflected generator requested in Q8 of *Exact Resonant Identities* [1].
2. The necessary and sufficient centered cancellation criterion for generalized harmonic polynomials requested in Q11 of the same report.
3. Simultaneous and hierarchical geometric collisions beyond the two-factor case, asked for in the periodic contact reports [4, 6].
4. The quartic directional layer and its cancellation map, requested in *Independent Orders and Exact Reductions*, Section 7, and the cubic harmonic report [2, 3].

These were posed as research questions. We do not recast them as named classical conjectures. The results below resolve the stated analytic and formal-algebraic components; the arithmetic reduction of the retained constants is a separate problem.

1.1 Basic functions and the meaning of a derivative

We use

$$\zeta(s, a) = \sum_{n=0}^{\infty} (n+a)^{-s}, \quad \text{Li}_s(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^s}$$

in their initial convergence domains, followed by the explicitly stated analytic continuation. The generalized Stieltjes functions are defined by

$$\zeta(1+u, x) = \frac{1}{u} + \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} \gamma_m(x) u^m. \quad (1.1)$$

In particular $\gamma_0(x) = -\psi(x)$, where $\psi = \Gamma'/\Gamma$. Write $\gamma = \gamma_0(1)$ and $\gamma_m = \gamma_m(1)$. A prime on ψ or γ_m means an argument derivative. A superscript $\zeta^{(j)}(s)$ means an order derivative of the Riemann zeta function. Multivariable subscripts on the Tornheim remainder mean partial derivatives, always evaluated at the point specified. These distinctions are essential at a spectral pole.

The generalized harmonic numbers are

$$H_n^{(r)} = \sum_{k=1}^n k^{-r}, \quad H_n = H_n^{(1)}.$$

All powers of positive real variables use the real logarithm. Thus, for complex λ , the expression $(\sin(\pi x)/\pi)^{2\lambda}$ on $0 < x < 1$ is unambiguous. Meromorphic identities are continued from a nonempty open convergence domain; a value at a crossing of polar divisors is never assigned by informal substitution.

1.2 Coordinate finite parts

For an endpoint expansion in real powers and integral logarithmic powers, the unit-coordinate finite part is the coefficient independent of both ϵ and $\log \epsilon$ in the cutoff integral. On the unit interval,

$$\text{FP} \int_0^1 f(x) dx = \text{CT}_{\epsilon \downarrow 0} \int_{\epsilon}^{1-\epsilon} f(x) dx. \quad (1.2)$$

The coordinates are x at zero and $1 - x$ at one. Independent endpoint cutoffs give the same additive constant. For example,

$$\text{FP} \int_0^1 x^{-1} \log^j x dx = 0, \quad \text{FP} \int_0^1 x^{-r} dx = \frac{1}{1-r} \quad (r \neq 1).$$

At periodic seams, the same convention is applied on each side in the unit argument coordinate. Geometric collision constants are then taken in a second parameter after this first finite part has been formed. The two operations can differ from first merging the factors.

All coefficient extractions below are from a germ proved holomorphic after its full prescribed subtraction. Numerical quadrature is used only to test identities already supplied with proofs. Exact finite linear algebra establishes a statement about its specified relation space, not arithmetic independence of the associated periods.

1.3 Relation to the literature

Hurwitz Fourier transforms, Beta and Gamma identities, and their applications to logarithmic integrals are classical [12, 13, 7]. Harmonic Stieltjes constants and log-sine connections were studied before this project [8, 9]. The Mellin continuation and Euler boundary relations for Tornheim zeta are likewise established tools [10]. The contributions claimed here are the particular completed generators, the exact all-polynomial criterion, the explicit cluster subtraction, and the higher directional reconstruction proved below. The source search was focused; it does not establish worldwide priority for every displayed specialization.

2 A completed complex-power sine transform

Put

$$\ell(x) = \log \frac{\sin \pi x}{\pi}, \quad L = \log(2\pi), \quad W_{\lambda}(x) = e^{2\lambda \ell(x)}. \quad (2.1)$$

The normalization by π makes the leading endpoint coefficient equal to one. It is useful both for coordinate finite parts and for keeping the later crossing subtraction unambiguous. Define

$$A(s, \lambda) = \int_0^1 \zeta(s, x) W_{\lambda}(x) dx, \quad C_0(\lambda) = \frac{\Gamma(1+2\lambda)}{\Gamma(1+\lambda)^2 (2\pi)^{2\lambda}}. \quad (2.2)$$

An initial domain sufficient for everything in the first theorem is $\Re \lambda > -1/2$, $\Re s < 0$. All subsequent values of A mean its meromorphic continuation.

2.1 Fourier coefficients and a polylogarithmic Euler integral

Theorem 2.1 (The complex-power transform). *For $\Re \lambda > -1/2$ the Fourier coefficients of W_λ are*

$$c_n(\lambda) = \int_0^1 W_\lambda(x) e^{-2\pi i n x} dx = \frac{(-1)^n \Gamma(1+2\lambda)}{(2\pi)^{2\lambda} \Gamma(1+\lambda+n) \Gamma(1+\lambda-n)}. \quad (2.3)$$

They satisfy $c_{-n} = c_n$, $c_0 = C_0$, and, for $n \geq 1$,

$$c_n(\lambda) = C_0(\lambda) \frac{(-\lambda)_n}{(1+\lambda)_n}. \quad (2.4)$$

Consequently,

$$A(s, \lambda) = B(s) \sum_{n=1}^{\infty} c_n(\lambda) n^{s-1}, \quad B(s) = 2\Gamma(1-s)(2\pi)^{s-1} \sin \frac{\pi s}{2}. \quad (2.5)$$

In the strip $-1/2 < \Re \lambda < 0$, $\Re s < 0$, a second representation is

$$\begin{aligned} A(s, \lambda) &= -\frac{2}{\pi} \Gamma(1-s)(2\pi)^{s-1-2\lambda} \sin \frac{\pi s}{2} \sin(\pi \lambda) \\ &\quad \times \int_0^1 t^{-\lambda-1} (1-t)^{2\lambda} \text{Li}_{1-s}(t) dt. \end{aligned} \quad (2.6)$$

Both formulas continue meromorphically wherever their two sides are defined by continuation.

Proof. Reflection about $1/2$ kills the sine coefficient. If $I_n = \int_0^\pi \sin^{2\lambda} t \cos(2nt) dt$, integration of $\frac{d}{dt} [\sin^{2\lambda+1} t \cos((2n+1)t)]$ gives

$$(\lambda - n)I_n + (\lambda + n + 1)I_{n+1} = 0.$$

The boundary term vanishes for $\Re \lambda > -1/2$. The beta integral gives $I_0 = \sqrt{\pi} \Gamma(\lambda+1/2)/\Gamma(\lambda+1)$. The recurrence and duplication formula prove (2.3)–(2.4). The apparent exceptions at integer parameters are read by continuity.

The Hurwitz Fourier expansion for $\Re s < 0$ is absolutely convergent. Its sine terms integrate to zero and its cosine terms give (2.5); see also [12, §25.11]. The decay $c_n = O(n^{-1-2\Re \lambda})$ shows that the resulting Dirichlet series itself converges absolutely for $\Re s < 1 + 2\Re \lambda$, away from exceptional parameters interpreted by continuity. Lastly, Gamma reflection gives

$$c_n(\lambda) = -\frac{\sin \pi \lambda}{\pi (2\pi)^{2\lambda}} \int_0^1 t^{n-\lambda-1} (1-t)^{2\lambda} dt.$$

Absolute convergence in the stated strip permits summation under the integral, producing (2.6). $\square$

These Fourier and beta mechanisms are classical. In particular, Espinosa and Moll treat Hurwitz transforms of trigonometric powers and the first log-sine transform [7, §§5,13]. The purpose of Theorem 2.1 here is to supply a common generator whose *completed germ at the joint resonance* can be specified at every Stieltjes and argument-derivative order.

For an integer $m \geq 1$ the Fourier expansion terminates:

$$A(s, m) = \frac{B(s)}{(2\pi)^{2m}} \sum_{n=1}^m (-1)^n \binom{2m}{m+n} n^{s-1}. \quad (2.7)$$

For example, comparison of the constant term at $s = 1$ gives the ordinary convergent identity

$$\int_0^1 \psi(x) W_m(x) dx = -C_0(m)(\gamma + L) + \frac{2}{(2\pi)^{2m}} \sum_{n=1}^m (-1)^n \binom{2m}{m+n} \log n. \quad (2.8)$$

Higher Laurent coefficients give the analogous all-index Stieltjes moments as finite Gamma and logarithmic expressions.

2.2 The complete crossing polar term

Let $P_p(u) = (1+u)_p$, with $P_0 = 1$, and define the entire-in- λ Taylor coefficients

$$\begin{aligned} d_j(\lambda) &= [x^{2j}] \left(\frac{\sin \pi x}{\pi x} \right)^{2\lambda} \\ &= [z^j] \exp \left(-2\lambda \sum_{r=1}^{\infty} \frac{\zeta(2r)}{r} z^r \right). \end{aligned} \quad (2.9)$$

Thus $d_0 = 1$, $d_1 = -2\lambda\zeta(2)$, and $d_2 = -\lambda\zeta(4) + 2\lambda^2\zeta(2)^2$.

Theorem 2.2 (All logarithmic Stieltjes moments). *For every integer $p \geq 0$ the expression*

$$\begin{aligned} \mathcal{H}_p(u, \lambda) &= (-1)^p P_p(u) A(1+p+u, \lambda) - \mathbf{1}_{p=0} \frac{C_0(\lambda)}{u} \\ &\quad - \mathbf{1}_{2|p} \frac{(-1)^p P_p(u) d_{p/2}(\lambda)}{2\lambda - u} \end{aligned} \quad (2.10)$$

extends holomorphically to a neighborhood of $(u, \lambda) = (0, 0)$. For all integers $m, k \geq 0$,

$$\boxed{\text{FP} \int_0^1 \gamma_m^{(p)}(x) \ell(x)^k dx = \frac{(-1)^m m! k!}{2^k} [u^m \lambda^k] \mathcal{H}_p(u, \lambda).} \quad (2.11)$$

The finite part is exactly the unit-coordinate convention (1.2).

Proof. The Hurwitz shift equation gives, near $x = 0$,

$$\partial_x^p \left(\zeta(1+u, x) - \frac{1}{u} \right) = (-1)^p P_p(u) x^{-1-p-u} + h_p(u, x), \quad (2.12)$$

where h_p is holomorphic jointly near $(u, x) = (0, 0)$. For $p = 0$ the subtraction $1/u$ is retained inside h_0 ; for $p > 0$ its derivative is zero. Moreover

$$W_\lambda(x) = x^{2\lambda} \sum_{j \geq 0} d_j(\lambda) x^{2j}.$$

Integration of the singular product from zero to a fixed $\delta \in (0, 1)$ produces the denominators $2\lambda - u + 2j - p$. Only $j = p/2$ can vanish at the origin, and it occurs only when p is even. Its full numerator is the one displayed in (2.10). Subtracting finitely many endpoint terms makes the remaining integral locally normally convergent; all the other denominators are bounded away from zero. The endpoint $x = 1$ has only the integrable analytic family $(1-x)^{2\lambda}$ times analytic functions and creates no divisor through $(0, 0)$. The pole $1/u$ of the original Hurwitz function contributes $C_0(\lambda)/u$ only when $p = 0$. This proves joint holomorphy.

For completeness, the local resonant integral is $N(u, \lambda)\delta^{2\lambda-u}/(2\lambda-u)$. After subtracting the entire quotient $N(u, \lambda)/(2\lambda-u)$, its Taylor coefficients are those of $N(u, \lambda)(\delta^{2\lambda-u} - 1)/(2\lambda-u)$. These are precisely the constants left by integrating the corresponding powers of $\log x$ in the coordinate x . All nonresonant terms have the same property by ordinary analytic continuation of $(\delta^a - \epsilon^a)/a$. Thus Taylor coefficients commute with the specified finite part after the full subtraction. Applying (1.1) and $W_\lambda = \sum_k 2^k \ell^k \lambda^k / k!$ proves (2.11). $\square$

Corollary 2.3. *For every $m, p \geq 0$,*

$$\text{FP} \int_0^1 \gamma_m^{(p)}(x) dx = 0.$$

Proof. The meromorphic identity $A(s, 0) = 0$ follows from (2.5), since all nonzero Fourier coefficients vanish. Also $d_j(0) = 0$ for $j > 0$. In (2.10) the two remaining terms for $p = 0$ cancel along $\lambda = 0$; for $p > 0$ each term vanishes there. Hence $\mathcal{H}_p(u, 0) = 0$ as a holomorphic germ. $\square$

Remark 2.4 (A removable limit need not equal a coordinate finite part). For an even integer $p \geq 2$, meromorphic continuation from a convergence domain gives

$$\lim_{\lambda \rightarrow 0} \left(\int_0^1 \psi^{(p)}(x) W_\lambda(x) dx \right)_{\text{cont}} = 2(p-1)! \zeta(p), \quad \text{FP} \int_0^1 \psi^{(p)}(x) dx = 0. \quad (2.13)$$

Indeed $d'_{p/2}(0) = -4\zeta(p)/p$, and the unsubtracted crossing quotient along $u = 0$ is $-p!d_{p/2}(\lambda)/(2\lambda)$ for the polygamma integral. This gives the first value, while Corollary 2.3 gives the second. For $p = 2$ the discrepancy is $2\zeta(2)$. The correction is to remove the *full analytic numerator*, including terms that vanish at the crossing. This is a normalization warning about a tempting extension, not a claim that the incoming completed-germ theorems make this error.

2.3 Cotangent powers and normalized antiderivatives

Let $K(x) = \pi \cot \pi x$. The identity $K^2 = W_{\lambda-1}/W_\lambda - \pi^2$ implies

$$\int_0^1 \zeta(s, x) K(x)^{2h} W_\lambda(x) dx = \sum_{j=0}^h \binom{h}{j} (-\pi^2)^{h-j} A(s, \lambda - j). \quad (2.14)$$

Since $W'_\mu = 2\mu K W_\mu$ and $\partial_x \zeta(s, x) = -s\zeta(s+1, x)$, integration by parts in an endpoint-convergent domain gives

$$\int_0^1 \zeta(s, x) K(x)^{2h+1} W_\lambda(x) dx = \sum_{j=0}^h \binom{h}{j} (-\pi^2)^{h-j} \frac{s}{2(\lambda-j)} A(s+1, \lambda-j). \quad (2.15)$$

These are meromorphic identities. A zero denominator on the right requires continuation and, for a coordinate finite part, the same local completion as above. The polynomial cotangent closure already proved in [1] is therefore embedded in a complex-power family rather than claimed afresh.

The symmetric Gamma primitive is particularly simple:

Proposition 2.5. *For $\Re \lambda > -1/2$,*

$$\int_0^1 \log \Gamma(x) W_\lambda(x) dx = -\frac{1}{4} C'_0(\lambda). \quad (2.16)$$

Consequently every $k \geq 0$ satisfies the ordinarily convergent identity

$$\int_0^1 \log \Gamma(x) \ell(x)^k dx = -\frac{C_0^{(k+1)}(0)}{2^{k+2}}. \quad (2.17)$$

Proof. Gamma reflection gives $\log \Gamma(x) + \log \Gamma(1-x) = -\ell(x)$. Multiply by the symmetric weight and integrate. The right side is $-C'_0(\lambda)/2$, while the left side is twice the desired integral. Differentiation is justified by endpoint domination on compact subsets of the stated half-plane. $\square$

For arbitrary Stieltjes index, the normalized antiderivative generator is

$$\mathcal{P}(u, x) = -\frac{\zeta(u, x) - \zeta(u, 1) + x - 1}{u}. \quad (2.18)$$

Its apparent singularity at $u = 0$ is removable because $\zeta(0, x) = 1/2 - x$. The Hurwitz derivative identity proves

$$\partial_x \mathcal{P}(u, x) = \zeta(1+u, x) - \frac{1}{u}, \quad \int_1^x \gamma_m(t) dt = (-1)^m m! [u^m] \mathcal{P}(u, x). \quad (2.19)$$

In particular $\mathcal{P}(0, x) = -\log \Gamma(x)$ by Lerch's formula. The normalization at $x = 1$ fixes the integration constant before any reflection or finite-part operation. Iterating ordinary integration of this holomorphic generator gives normalized higher primitives.

2.4 Harmonic jets of the weight

Proposition 2.6. *For every integer $n \geq 1$ the germ of its Fourier coefficient is*

$$c_n(\lambda) = -\frac{\lambda}{n} \exp \left(-2L\lambda + \sum_{j \geq 2} \frac{(-1)^j (2^j - 2) \zeta(j)}{j} \lambda^j - \sum_{j \geq 1} \frac{H_{n-1}^{(j)} + (-1)^{j+1} H_n^{(j)}}{j} \lambda^j \right). \quad (2.20)$$

Thus every logarithmic sine moment is generated by finite polynomials in generalized harmonic numbers and ordinary zeta constants.

Proof. Factor (2.4) as

$$c_n = -\frac{\lambda}{n} C_0(\lambda) \frac{\prod_{j=1}^{n-1} (1 - \lambda/j)}{\prod_{j=1}^n (1 + \lambda/j)}.$$

Expanding the logarithm of each finite product and using the Taylor series of $\log \Gamma(1 + \lambda)$ gives (2.20). Each fixed coefficient requires only finitely many terms. Initially the differentiated Dirichlet series converges normally in a smaller half-plane; its extension is then supplied by Theorem 2.2. $\square$

The first two terms illustrate the inclusive harmonic convention:

$$c_n(\lambda) = -\frac{\lambda}{n} + \lambda^2 \left(\frac{2(L + H_n)}{n} - \frac{1}{n^2} \right) + O(\lambda^3). \quad (2.21)$$

More explicitly, if $J_k(s) = \int_0^1 \zeta(s, x) \ell(x)^k dx$, then in its initial domain

$$J_k(s) = \frac{k!}{2^k} B(s) \sum_{n \geq 1} [\lambda^k] c_n(\lambda) n^{s-1}. \quad (2.22)$$

This is a finite Bell-polynomial construction at each fixed k , not an assertion that all resulting harmonic constants have already been reduced to ordinary ones.

2.5 The first two logarithmic moments and an exact cancellation

For $k = 1$, the classical transform becomes

$$\begin{aligned} J_1(s) &= -\Gamma(1-s)(2\pi)^{s-1} \sin \frac{\pi s}{2} \zeta(2-s) \\ &= \frac{\pi \cot(\pi(s-1)/2)}{s-1} \zeta(s-1). \end{aligned} \quad (2.23)$$

The second expression follows from the Riemann functional equation. The completed germ is

$$J_1(1+u) + \frac{1}{u^2} + \frac{L}{u}. \quad (2.24)$$

Its u^m coefficient, multiplied by $(-1)^m m!$, is $\text{FP} \int \gamma_m \ell$. In particular,

$$\text{FP} \int_0^1 \gamma_0(x) \ell(x) dx = \zeta''(0) + \frac{\zeta(2)}{2}. \quad (2.25)$$

This also cross-checks the log-Gamma-cotangent specialization in [1]; neither (2.23) nor that specialization is claimed to be new.

Write the *inclusive* harmonic zeta function as

$$\zeta_H(v) = \sum_{n \geq 1} \frac{H_n}{n^v} = \frac{1}{(v-1)^2} + \frac{\gamma}{v-1} + \sum_{j \geq 0} \frac{(-1)^j \gamma_H(j)}{j!} (v-1)^j. \quad (2.26)$$

Here $\gamma_H(0) = (\gamma^2 + \zeta(2))/2$; this normalization and its Laurent theory are established in [8, 9].

Theorem 2.7 (The sine-square bridge). *One has*

$$J_2(s) = B(s) \left\{ L\zeta(2-s) + \zeta_H(2-s) - \frac{1}{2}\zeta(3-s) \right\}. \quad (2.27)$$

Its completion at $s = 1 + u$ is

$$J_2(1+u) + \frac{2}{u^3} - \frac{L^2 + \zeta(2)/2}{u}. \quad (2.28)$$

In particular,

$$\begin{aligned} \text{FP} \int_0^1 \psi(x) \ell(x)^2 dx &= 2\gamma_H(1) + \zeta'(2) + 2L\gamma_1 + \frac{\gamma^3}{3} + \gamma^2 L \\ &\quad - \frac{2L^3}{3} + \frac{2\zeta(3)}{3}. \end{aligned} \quad (2.29)$$

Combining this with the already proved cubic harmonic bridge yields the ordinary-constant identity

$$\boxed{\begin{aligned} \text{FP} \int_0^1 \{ \psi(x)^3 - 3\psi(x)\ell(x)^2 \} dx &= 3\zeta(2) + 3\zeta'(2) - 6(\gamma + L)\gamma_1 \\ &\quad - \gamma^3 - 3\gamma^2 L + 2L^3 - 2\zeta(3). \end{aligned}} \quad (2.30)$$

Proof. Equation (2.27) is the second coefficient of (2.21) in (2.22). The polar terms of (2.28) also follow directly by taking the second λ coefficient of $C_0(\lambda)/u + 1/(2\lambda - u)$ in Theorem 2.2. To make the constant calculation explicit, set $a = \gamma + L$. Then

$$B(1+u) = -\frac{2}{u} \exp \left(au - \frac{\zeta(2)}{4} u^2 + \frac{\zeta(3)}{3} u^3 + O(u^4) \right),$$

and the braces in (2.27) equal

$$\frac{1}{u^2} - \frac{a}{u} + L\gamma + \frac{\gamma^2}{2} + u \left(L\gamma_1 + \gamma_H(1) + \frac{\zeta'(2)}{2} \right) + O(u^2).$$

Multiplication gives the negative of (2.29), because $\gamma_0 = -\psi$.

The incoming cubic report [3] proves, in precisely this coordinate convention,

$$\text{FP} \int_0^1 \psi(x)^3 dx = 3\zeta(2) + 6\gamma_H(1) + 6\zeta'(2) - 6\gamma\gamma_1. \quad (2.31)$$

Subtracting three times (2.29) cancels $\gamma_H(1)$ identically and gives (2.30). This use of the cubic bridge is explicit: we prove the new cancellation, not a new proof claim for the inherited bridge. $\square$

The individual quantities are, for orientation,

$$\begin{aligned} \text{FP} \int_0^1 \psi(x)^3 dx &= 1.9662627343915343406643753384116314\dots, \\ \text{FP} \int_0^1 \psi(x)\ell(x)^2 dx &= -3.0645068706816349567044046185265270\dots \end{aligned}$$

Their combination in (2.30) is 11.1597833464364392107775891939912123.... These decimals come from independent endpoint-subtracted quadratures and harmonic-tail evaluation in the supplied check script. They are diagnostics; the coefficient identities and the inherited theorem are the proof. No reduction of $\gamma_H(1)$ itself to ordinary constants is inferred from its disappearance in this particular combination.

3 A complete criterion for centered harmonic pole cancellation

Question 11 of the incoming Exact Resonant Identities report [1] asks which polynomials in generalized harmonic numbers have no principal part at any nonpositive even spectral integer when the outer shift is $1/2$. The criterion below is finite and necessary as well as sufficient. The necessity is the substantive point: parity of the individual asymptotic expansions alone supplies only one implication.

For a fixed $R \geq 1$, write

$$H_n^{(r)} = \sum_{k=1}^n k^{-r}, \quad H_n^{(1)} = H_n, \quad D_P(s) = \sum_{n=0}^{\infty} \frac{P(H_n^{(1)}, \dots, H_n^{(R)})}{(n + 1/2)^s}, \quad P \in \mathbb{C}[X_1, \dots, X_R].$$

The defining series is absolutely convergent for $\Re s > 1$. As usual $H_0^{(r)} = 0$, so the finite $n = 0$ term is retained. Define an involution ι on the polynomial ring by

$$\iota(X_{2k}) = 2\zeta(2k) - X_{2k}, \quad \iota(X_{2k+1}) = X_{2k+1}, \quad \iota(X_1) = X_1. \quad (3.1)$$

Only variables whose indices are at most R are used.

Theorem 3.1 (Necessary and sufficient centered cancellation). *The function D_P has a meromorphic continuation to $\mathbb{C}$. It is holomorphic at every point $s = 0, -2, -4, \dots$ if and only if*

$$\iota(P) = P. \quad (3.2)$$

Equivalently, put $Y_{2k} = X_{2k} - \zeta(2k)$. After expressing P in the variables $X_1, X_3, X_5, \dots$ and $Y_2, Y_4, Y_6, \dots$, every monomial that occurs must have even total degree in the Y variables. The invariant algebra is therefore

$$\mathbb{C}[X_1, X_3, X_5, \dots, Y_{2i}Y_{2j} \ (1 \leq i \leq j \leq \lfloor R/2 \rfloor)]. \quad (3.3)$$

This is a generating description, not a claim that the displayed quadratic generators are algebraically independent.

For example, every polynomial in odd-order harmonic numbers has the cancellation, as does every product of an even number of even-order tails. Thus the numerators

$$H_n^{(3)}, \quad H_n^2 H_n^{(5)}, \quad (H_n^{(2)} - \zeta(2))^2, \quad (H_n^{(2)} - \zeta(2))(H_n^{(4)} - \zeta(4))$$

all satisfy the criterion. The two even-order factors need not be equal. The raw numerator $(H_n^{(2)})^2$ fails: its simple-pole coefficient at $s = 0$ is $-2\zeta(2)$.

3.1 Bernoulli coefficients and the principal parts

Use $x = n + 1/2$, $z = x^{-1}$, and the formal symbol $\mathcal{L} = \log x$. Define

$$h_1(z, \mathcal{L}) = \mathcal{L} + \gamma - \sum_{k \geq 1} \frac{B_{2k}(1/2)}{2k} z^{2k}, \quad (3.4)$$

$$h_r(z) = \zeta(r) - \frac{z^{r-1}}{r-1} - \sum_{k \geq 1} \frac{B_{2k}(1/2)}{(2k)!} (r)_{2k-1} z^{r+2k-1}, \quad r \geq 2. \quad (3.5)$$

These are formal asymptotic series; no convergence of the infinite Bernoulli expansions is asserted. Each finite truncation is the usual shifted polygamma or Hurwitz-zeta asymptotic expansion [14, 12], with a remainder sufficient for termwise Dirichlet subtraction. Expand

$$P(h_1, h_2, \dots, h_R) = \sum_{j \geq 0} A_j(\mathcal{L}) z^j, \quad A_j \in \mathbb{C}[\mathcal{L}]. \quad (3.6)$$

Every fixed A_j is a finite calculation. The degree in $\mathcal{L}$ is bounded by $\deg_{X_1} P$.

Proposition 3.2 (All principal coefficients). *At $s = 1 - j$, $j \geq 0$, the entire Laurent principal part of D_P is*

$$\sum_{\ell \geq 0} \frac{\ell! [\mathcal{L}^\ell] A_j(\mathcal{L})}{(s + j - 1)^{\ell+1}}. \quad (3.7)$$

Consequently D_P is regular at all nonpositive even integers exactly when $A_{2m+1} = 0$ for every $m \geq 0$.

Proof. Subtract finitely many terms of (3.6) from the defining series. Each subtracted term sums to $(-1)^\ell \partial_s^\ell \zeta(s + j, 1/2)$ times its coefficient. The finite asymptotic remainder gives a normally convergent series in a successively larger left half-plane. These formulas agree on overlaps and continue D_P meromorphically. The only principal coefficient of $(-1)^\ell \partial_s^\ell \zeta(s + j, 1/2)$ is $\ell!/(s + j - 1)^{\ell+1}$, proving the claim. Distinct logarithmic degrees give distinct Laurent powers, so none is lost by merely checking the residue. $\square$

Here are the first three finite Bernoulli tests. Let ∂_r denote polynomial differentiation with respect to X_r , omitting indices larger than R , and define commuting operators

$$\begin{aligned} A &= -\partial_2, & B &= \frac{\partial_1}{24} - \frac{\partial_3}{2}, & C &= \frac{\partial_2}{12} - \frac{\partial_4}{3}, \\ D &= -\frac{7\partial_1}{960} + \frac{\partial_3}{8} - \frac{\partial_5}{4}, & E &= -\frac{7\partial_2}{240} + \frac{\partial_4}{6} - \frac{\partial_6}{5}. \end{aligned}$$

Evaluate the following expressions at $\mathbf{b}(\mathcal{L}) = (\mathcal{L} + \gamma, \zeta(2), \dots, \zeta(R))$:

$$\begin{aligned} A_1(\mathcal{L}) &= (AP)(\mathbf{b}(\mathcal{L})), \\ A_3(\mathcal{L}) &= \left((C + AB + A^3/6)P \right)(\mathbf{b}(\mathcal{L})), \\ A_5(\mathcal{L}) &= \left((E + AD + BC + (A^2C + AB^2)/2 + A^3B/6 + A^5/120)P \right)(\mathbf{b}(\mathcal{L})). \end{aligned} \tag{3.8}$$

They follow by expanding the finite translation operator $\exp(zA + z^2B + z^3C + z^4D + z^5E + \dots)$. For example,

$$A_3 = \left(\frac{P_2}{12} - \frac{P_4}{3} - \frac{P_{12}}{24} + \frac{P_{23}}{2} - \frac{P_{222}}{6} \right)(\mathbf{b}(\mathcal{L})).$$

The rational-coefficient polynomial $P = X_2^3 - \frac{15}{2}X_2X_4$ satisfies $A_1 = 0$, because $\zeta(4) = 2\zeta(2)^2/5$, but $A_3 = \frac{5}{2}\zeta(2) - 1 \neq 0$. Thus its Dirichlet series is regular at zero and has a simple pole at -2 . More generally $P = X_{2M+2} - \zeta(2M+2)$ passes all earlier tests and first has a pole at $s = -2M$, with residue $-1/(2M+1)$. Finite lists of tested spectral centers cannot replace the polynomial criterion when R is unrestricted.

3.2 Why the parity criterion is necessary

The following elementary difference-algebra lemma is the relevant form of the classical additive-difference mechanism behind Hölder's theorem; compare the general theory in [11]. It is proved here, so the formal-series application does not rely on a converse assertion about analytic asymptotic expansions.

Lemma 3.3 (A polynomial relation forces rational telescoping). *Let $K = \mathbb{C}(x)$ with $\sigma x = x + 1$, and let E be a field extension carrying an extension of σ . Suppose $f_1, \dots, f_d \in E$ satisfy $\sigma f_i = f_i + a_i$, $a_i \in K$. If the f_i are algebraically dependent over K , then there exist constants $c_i \in \mathbb{C}$, not all zero, and $b \in K$ such that*

$$\sum_i c_i a_i = \sigma b - b. \tag{3.9}$$

Proof. Choose a nonzero polynomial relation $Q(\mathbf{f}) = 0$ of least total degree d_0 . Among such relations choose one with the fewest nonzero monomials of degree d_0 , and divide by one of its leading coefficients. That coefficient is now one. The polynomial

$$\sigma Q(\mathbf{X} + \mathbf{a}) - Q(\mathbf{X})$$

also vanishes at $\mathbf{f}$. Its chosen leading monomial cancels, and no new degree- d_0 monomial is created. Minimality first of degree and then of leading support forces this difference to be the zero polynomial. If Q_{d_0} is the highest homogeneous part, then $\sigma Q_{d_0} = Q_{d_0}$. Every rational periodic coefficient is constant, so $Q_{d_0} \in \mathbb{C}[\mathbf{X}]$. Comparison in degree $d_0 - 1$ gives

$$\sigma Q_{d_0-1} - Q_{d_0-1} = - \sum_i a_i \partial_i Q_{d_0}.$$

At least one coefficient of the gradient on the right is a nonzero vector in $\mathbb{C}^d$, since $d_0 \geq 1$ and the characteristic is zero. Taking that monomial coefficient and reversing the sign of the rational coefficient on the left proves (3.9). $\square$

Lemma 3.4 (Independence of the formal centered harmonic series). *For every finite R , the formal series $h_1, \dots, h_R$ in (3.4)–(3.5) are algebraically independent over $\mathbb{C}(x)$, where $x = z^{-1}$ and $\mathcal{L}$ is a formal logarithm.*

Proof. Work in $E = \mathbb{C}((z))(\mathcal{L})$. Extend translation by

$$\sigma z = \frac{z}{1+z}, \quad \sigma \mathcal{L} = \mathcal{L} + \log(1+z),$$

and use the derivation $\mathcal{D}z = -z^2$, $\mathcal{D}\mathcal{L} = z$. These operations commute. Put $f = h_1 - \gamma$. The Bernoulli expansion and the digamma recurrence give the formal identity

$$\sigma f - f = \frac{1}{x+1/2}, \quad \sigma \mathcal{D}^j f - \mathcal{D}^j f = \frac{(-1)^j j!}{(x+1/2)^{j+1}} \quad (j \geq 0).$$

It can equally be checked coefficientwise from the Bernoulli generating function. A nonzero constant linear combination of the right sides has exactly one finite pole, at $-1/2$.

No nonzero rational difference $b(x+1) - b(x)$ has just one finite pole. Indeed partition the finite poles of b into classes modulo $\mathbb{Z}$. In any nonempty class, the extreme poles of b give two distinct, uncanceled extreme poles of its difference. If b has no finite poles, it is polynomial and its difference is polynomial. Lemma 3.3 therefore proves algebraic independence of $f, \mathcal{D}f, \dots, \mathcal{D}^{R-1}f$. Finally

$$h_1 = f + \gamma, \quad h_r = \zeta(r) - \frac{(-1)^r}{(r-1)!} \mathcal{D}^{r-1}f \quad (r \geq 2)$$

is an invertible affine change of these variables. $\square$

Proof of Theorem 3.1. Let τ act on formal expansions by $z \mapsto -z$, $\mathcal{L} \mapsto \mathcal{L}$. Equations (3.4)–(3.5) show $\tau h_r = \iota(h_r)$: the even harmonic tails change sign and the odd harmonic variables do not. By Proposition 3.2, absence of every nonpositive even principal part is equivalent to $\tau P(\mathbf{h}) = P(\mathbf{h})$, hence to $(\iota P - P)(\mathbf{h}) = 0$. Lemma 3.4 makes this equivalent to the polynomial identity $\iota P = P$. In centered variables Y_{2k} the involution changes all signs simultaneously. Its fixed monomials have even total Y -degree, and each such monomial factors into quadratic pairs. This proves (3.3). $\square$

Corollary 3.5 (Algebraic coefficients). *If P has algebraic-number coefficients, the criterion holds if and only if P is independent of every even-index variable.*

Proof. Write $\zeta(2k) = c_k \pi^{2k}$ with nonzero $c_k \in \mathbb{Q}$. Since π^2 is transcendental, invariance of a polynomial over $\overline{\mathbb{Q}}$ under (3.1) implies its invariance under the polynomial reflection $X_{2k} \mapsto 2c_k T^k - X_{2k}$ for every indeterminate T . Composing two such reflections gives invariance under translations by $2(c_k T^k - c_k U^k)_k$. Differentiating in T and then setting $T = U$ yields

$$\sum_k k c_k U^{k-1} \partial_{2k} P = 0.$$

Distinct powers of U force each $\partial_{2k} P$ to vanish. The converse follows directly from Theorem 3.1. $\square$

This corollary uses only the established transcendence of π . The main theorem makes no assertion about algebraic independence of evaluated zeta or Stieltjes constants.

3.3 All centered even moments of a trigamma square

The invariant numerator $(H_n^{(2)} - \zeta(2))^2$ admits a particularly short all-degree value formula. Put

$$T_n = \zeta(2, n+1) = \psi'(n+1), \quad V_M = D_{(X_2 - \zeta(2))^2}(-2M), \quad M \geq 0.$$

Every point in this definition is regular by Theorem 3.1.

Theorem 3.6 (Finite Bernoulli formula for all even moments). *For every $M \geq 0$,*

$$V_M = 3B_{2M}(1/2)\zeta(3) - \frac{1}{2(2M+1)} \sum_{\ell=1}^M \binom{2M+1}{2M-2\ell} B_{2M-2\ell}(1/2)(2\ell-1)B_{2\ell-2}. \quad (3.10)$$

The sum is empty at $M = 0$. In particular all these values belong to $\mathbb{Q} + \mathbb{Q}\zeta(3)$.

Proof. The polynomial partial sum is

$$S_M(k) = \sum_{n=0}^{k-1} (n+1/2)^{2M} = \frac{B_{2M+1}(k+1/2)}{2M+1} = \sum_{\ell=0}^M s_{M,\ell} k^{2\ell+1},$$

where $s_{M,\ell} = \binom{2M+1}{2M-2\ell} B_{2M-2\ell}(1/2)/(2M+1)$. Since $T_{k-1} - T_k = k^{-2}$, finite summation by parts gives

$$\sum_{n=0}^{N-1} (n+1/2)^{2M} T_n^2 = \sum_{\ell=0}^M s_{M,\ell} E_{2\ell+1}(N),$$

with

$$E_q(N) = N^q T_N^2 + 2 \sum_{k=1}^N k^{q-2} T_{k-1} - \sum_{k=1}^N k^{q-4}.$$

At $q = 1$ the limit is $3\zeta(3)$: the classical Euler sum [9] $\sum_{k \geq 1} H_k/k^2 = 2\zeta(3)$ implies $\sum_{k \geq 1} T_{k-1}/k = 2\zeta(3)$, by interchanging a positive double sum. The remaining diagonal term is $\zeta(3)$.

For odd $q \geq 3$, let $P_{q-2}(N) = \sum_{k=1}^N k^{q-2}$. Interchanging a finite sum gives

$$\sum_{k=1}^N k^{q-2} T_{k-1} = \sum_{j=1}^N \frac{P_{q-2}(j)}{j^2} + P_{q-2}(N) T_N.$$

In Faulhaber's polynomial the half-leading term cancels $\sum k^{q-4}$ from E_q . All other finite power sums are polynomials in N with zero constant coefficient. Hence the constant term of the expansion of $E_q(N)$ at infinity is the constant of $N^q T_N^2 + 2P_{q-2}(N) T_N$. Use

$$T_N = N^{-1} - \frac{1}{2}N^{-2} + \sum_{j \geq 1} B_{2j} N^{-2j-1}.$$

Only finitely many terms are needed. The first product contributes $-B_{q-3}$ and the second contributes $B_{q-3} - (q-2)B_{q-3}/2$. Their sum is $-(q-2)B_{q-3}/2$.

It remains to identify this cutoff constant with the stated spectral value. The centered expansion of T_n^2 contains only even powers of $(n+1/2)^{-1}$. Subtract from $(n+1/2)^{2M} T_n^2$ all its nondecaying even powers. The residual is $O(n^{-2})$ and is ordinarily summable. Each subtracted power has spectral value $\zeta(-2j, 1/2) = 0$, and its finite partial sum is an odd polynomial in N with zero constant. Thus the residual sum, the cutoff constant in N , and V_M coincide. Combining the coefficients $s_{M,\ell}$ proves (3.10). $\square$

The first four identities can be written as ordinary convergent bracketed series, with $x_n = n + 1/2$:

$$\begin{aligned}
\sum_{n \geq 0} T_n^2 &= 3\zeta(3), \\
\sum_{n \geq 0} \{x_n^2 T_n^2 - 1\} &= -\frac{1}{6} - \frac{1}{4}\zeta(3), \\
\sum_{n \geq 0} \{x_n^4 T_n^2 - x_n^2 + \frac{1}{6}\} &= \frac{1}{30} + \frac{7}{80}\zeta(3), \\
\sum_{n \geq 0} \{x_n^6 T_n^2 - x_n^4 + \frac{1}{6}x_n^2 - \frac{47}{720}\} &= \frac{1}{672} - \frac{31}{448}\zeta(3).
\end{aligned} \tag{3.11}$$

Each bracket is $O(n^{-2})$. These statements do not assign an ordinary sum to any of the divergent unbracketed expressions.

4 Geometric collisions at Stieltjes index zero

The periodic contact reports [4, 5, 6] develop two-factor and spectral completions. Here the shifts themselves move toward collision. The result covers every finite number of factors and every argument-derivative order at Stieltjes index zero.

Set $f(x) = \gamma_0(x) = -\psi(x)$ and $h(x) = f(1+x)$. The Hurwitz shift identity gives the exact local decomposition

$$f(\{x\}) = \frac{\mathbf{1}_{x>0}}{x} + h(x), \quad -1 < x < 1, \quad x \neq 0. \tag{4.1}$$

Here the negative half of the formula uses $\{x\} = 1+x$. The germ h is holomorphic on $|x| < 1$, with

$$h(x) = \gamma + \sum_{n \geq 1} (-1)^n \zeta(n+1) x^n. \tag{4.2}$$

The finite-part convention is fixed throughout: at each right-hand singular point a remove $(a, a + \eta)$ and take the constant term in the *ambient coordinate* η . Independent cutoffs are used at distinct points. No collision-dependent rescaling of these coordinates is made.

For distinct sufficiently small real shifts $s_1, \dots, s_d$, define

$$C_d(\mathbf{s}) = \text{FP} \int_{\mathbb{R}/\mathbb{Z}} \prod_{i=1}^d f(\{x + s_i\}) dx, \quad Q_d = \text{FP} \int_0^1 f(x)^d dx. \tag{4.3}$$

For a nonempty subset $S \subseteq \{1, \dots, d\}$, let $m(S)$ denote its index with smallest shift. It is a fixed index when an ordering chamber is fixed.

Theorem 4.1 (Complete local counterterm for any number of factors). *Define*

$$T_d(\mathbf{s}) = - \sum_{\substack{S \subseteq \{1, \dots, d\} \\ |S| \geq 2}} \sum_{i \in S \setminus \{m(S)\}} \frac{\prod_{j \notin S} h(s_j - s_i)}{\prod_{k \in S \setminus \{i\}} (s_k - s_i)} \log(s_i - s_{m(S)}). \tag{4.4}$$

On each fixed ordering chamber, $C_d(\mathbf{s}) - T_d(\mathbf{s})$ is the restriction of a real analytic function on a full neighborhood of $\mathbf{s} = 0$. Its value at total collision is Q_d . Consequently

$$C_d(\mathbf{s}) - T_d(\mathbf{s}) \longrightarrow Q_d \quad (\mathbf{s} \longrightarrow 0) \tag{4.5}$$

along every path remaining in that chamber, without a lower bound on the ratios of the gaps.

Proof. Choose $\delta > 0$ small enough that all required germs are holomorphic on $|z| < 3\delta$, and restrict the shifts to $|s_i| < \delta/8$. Use the circle coordinate $-\delta < x < 1 - \delta$. Outside $(-\delta, \delta)$ the entire integrand is jointly analytic in the shifts, and its integral is analytic. On $(-\delta, \delta)$, expand the exact product of (4.1) by the subset S of factors contributing their pole. The empty subset is analytic. For a nonempty S , the product of the indicator functions is $\mathbf{1}_{x > -s_{m(S)}}$, and the contribution is the finite part of

$$\int_{-s_{m(S)}}^{\delta} \frac{q_S(x; \mathbf{s})}{D_S(x; \mathbf{s})} dx, \quad q_S(x; \mathbf{s}) = \prod_{j \notin S} h(x + s_j), \quad D_S(x; \mathbf{s}) = \prod_{i \in S} (x + s_i). \quad (4.6)$$

Let P_S be the interpolation polynomial of degree less than $|S|$ which agrees with q_S at the nodes $-s_i$, $i \in S$. Both P_S and $R_S = (q_S - P_S)/D_S$ extend holomorphically when nodes coalesce. One explicit justification is the fixed-contour expression

$$P_S(x; \mathbf{s}) = \frac{1}{2\pi i} \oint_{|z|=2\delta} \frac{q_S(z; \mathbf{s})}{D_S(z; \mathbf{s})} \frac{D_S(z; \mathbf{s}) - D_S(x; \mathbf{s})}{z - x} dz. \quad (4.7)$$

Its integrand is holomorphic in the parameters and the quotient of polynomials is a polynomial in x of the required degree. Cauchy's formula gives the jointly holomorphic remainder

$$R_S(x; \mathbf{s}) = \frac{1}{2\pi i} \oint_{|z|=2\delta} \frac{q_S(z; \mathbf{s})}{D_S(z; \mathbf{s})(z - x)} dz, \quad |x| < 2\delta.$$

For distinct nodes the rational principal part is

$$\frac{P_S(x; \mathbf{s})}{D_S(x; \mathbf{s})} = \sum_{i \in S} \frac{b_{S,i}(\mathbf{s})}{x + s_i}, \quad b_{S,i}(\mathbf{s}) = \frac{q_S(-s_i; \mathbf{s})}{\prod_{k \in S \setminus \{i\}} (s_k - s_i)}.$$

Its finite-part integral in (4.6) equals

$$\sum_{i \in S} b_{S,i} \log(\delta + s_i) - \sum_{i \in S \setminus \{m(S)\}} b_{S,i} \log(s_i - s_{m(S)}). \quad (4.8)$$

At the minimal node the discarded primitive is $b_{S,m(S)} \log \eta$, whose constant term is zero under the specified coordinate convention. The second sum in (4.8) is precisely the corresponding summand of (4.4). Although the individual coefficients $b_{S,i}$ have poles at coincident nodes, the first sum has the holomorphic form

$$\frac{1}{2\pi i} \oint_{|z|=\delta/2} \frac{q_S(z; \mathbf{s})}{D_S(z; \mathbf{s})} \text{Log}(\delta - z) dz. \quad (4.9)$$

The logarithm is the holomorphic branch equal to $\log \delta$ at $z = 0$. This contour is deliberately smaller than the interpolation contour in (4.7). The integral of R_S from $-s_{m(S)}$ to δ is also holomorphic, since its lower endpoint is a fixed coordinate function on the chosen chamber.

At $\mathbf{s} = 0$, $D_S = x^{|S|}$ and P_S is the Taylor polynomial of q_S through degree $|S| - 1$. Formula (4.9) is exactly the finite-part integral of $P_S/x^{|S|}$ from 0 to δ ; the remainder gives the ordinary integral of the analytic part. Summing the subsets therefore gives the same local finite part as the directly merged product. The outer integral supplies its remaining part, proving that the analytic remainder has value Q_d . $\square$

Remark 4.2 (Meaning of the chamber qualification). The extension asserted above is an extension *from each chamber*. Different chambers choose different lower-endpoint indices in (4.6); no assertion that the resulting analytic germs agree off the total diagonal is needed. All their values at total collision are the prescribed merged finite part. In particular the theorem proves a path-independent limit only after subtracting the full T_d , including any finite constant it contributes on the chosen path.

4.1 Every argument-derivative order

For nonnegative integers $r_1, \dots, r_d$, put

$$d_i = r_i + 1, \quad A_i = (-1)^{r_i} r_i!, \quad h_i(x) = \gamma_0^{(r_i)}(1+x).$$

Ordinary differentiation away from the integer endpoints gives

$$\gamma_0^{(r_i)}(\{x\}) = A_i \mathbf{1}_{x>0} x^{-d_i} + h_i(x).$$

Define $C_{\mathbf{r}}(\mathbf{s})$ and $Q_{\mathbf{r}}$ by replacing the factors in (4.3) by the indicated argument derivatives. For a nonempty S , set

$$A_S = \prod_{i \in S} A_i, \quad q_S(x; \mathbf{s}) = \prod_{j \notin S} h_j(x + s_j), \quad D_S(x; \mathbf{s}) = \prod_{i \in S} (x + s_i)^{d_i}.$$

For $i \in S$ and $1 \leq q \leq d_i$, let

$$B_{S,i,q}(\mathbf{s}) = \frac{1}{(d_i - q)!} \left(\frac{d}{dx} \right)^{d_i - q} \frac{q_S(x; \mathbf{s})}{\prod_{k \in S \setminus \{i\}} (x + s_k)^{d_k}} \Big|_{x = -s_i}. \quad (4.10)$$

Theorem 4.3 (Repeated-pole collision formula). *Let*

$$T_{\mathbf{r}}(\mathbf{s}) = \sum_{\substack{S \subseteq \{1, \dots, d\} \\ |S| \geq 2}} A_S \sum_{i \in S \setminus \{m(S)\}} \left\{ -B_{S,i,1} \log(s_i - s_{m(S)}) + \sum_{q=2}^{d_i} \frac{B_{S,i,q}}{q-1} (s_i - s_{m(S)})^{1-q} \right\}. \quad (4.11)$$

Then $C_{\mathbf{r}} - T_{\mathbf{r}}$ has a real analytic extension from each ordering chamber through total collision, with value $Q_{\mathbf{r}}$ there.

Proof. Repeat the preceding proof with Hermite interpolation of multiplicity d_i at $-s_i$. Formula (4.7), with the new D_S , proves holomorphy of the interpolation polynomial and its remainder, including coalescing nodes. The principal part is $\sum_{i \in S} \sum_{q=1}^{d_i} B_{S,i,q}(x + s_i)^{-q}$. Its upper primitive is again (4.9), multiplied by A_S , since

$$\frac{1}{(q-1)!} \frac{d^{q-1}}{dz^{q-1}} \text{Log}(\delta - z) = -\frac{(\delta - z)^{1-q}}{q-1}, \quad q \geq 2.$$

At a nonminimal node, subtracting the lower primitive gives exactly the bracket in (4.11). At the minimal node each lower primitive is a pure power of η or a multiple of $\log \eta$, and hence has zero constant term. The same coalescence argument identifies the value of the analytic remainder with $Q_{\mathbf{r}}$. $\square$

This proof differentiates only ordinary local analytic functions in (4.10). Differentiating an already extended periodic distribution can generate contact terms, and does not supply a substitute for the direct repeated-pole proof. For example, with two factors and shifts $(0, a)$, the formula gives

$$T_{(1,0)}(0, a) = \frac{\log a}{a^2}, \quad T_{(0,1)}(0, a) = \frac{1 - \log a}{a^2}.$$

Both analytic remainders tend to $Q_{(1,0)} = Q_{(0,1)} = \zeta(2)$, which also follows by integrating $ff' = (f^2)'/2$ and taking the endpoint finite part. The separated integrals satisfy the exact identity

$$C_{(1,0)}(0, a) + C_{(0,1)}(0, a) = \frac{\pi^2}{\sin^2(\pi a)}, \quad 0 < a < 1. \quad (4.12)$$

Indeed, apply the fundamental theorem of calculus to the product $f(x)f(\{x + a\})$ on the two nonsingular intervals. At 0 the left boundary value minus the constant in the right Laurent expansion is $-f'(a)$; at $1 - a$ it is $-f'(1 - a)$. Taking the finite parts and using trigamma reflection gives $-f'(a) - f'(1 - a) = \psi'(a) + \psi'(1 - a) = \pi^2/\sin^2(\pi a)$. Thus (4.12) records the boundary correction required by the specified finite part. The sum of the two local counterterms is a^{-2} , and the sum of their limiting remainders is $2\zeta(2) = \pi^2/3$, in agreement with the Taylor expansion of its right side.

4.2 Fixed rates and a finite zeta–logarithm expression

For $s_i = a_i\epsilon$ with fixed distinct real rates, define $m(S) = \operatorname{argmin}_{i \in S} a_i$. Coefficient extraction from (4.4) gives

$$\begin{aligned} \text{CT}_{\epsilon \downarrow 0} C_d(\epsilon \mathbf{a}) = Q_d - \sum_{\substack{S \subseteq \{1, \dots, d\} \\ |S| \geq 2}} \sum_{i \in S \setminus \{m(S)\}} \frac{\log(a_i - a_{m(S)})}{\prod_{k \in S \setminus \{i\}} (a_k - a_i)} \\ \cdot [\epsilon^{|S|-1}] \prod_{j \notin S} h(\epsilon(a_j - a_i)). \end{aligned} \quad (4.13)$$

Here the constant-term operation discards positive powers of $\log \epsilon$ even when their power of ϵ is zero. Only finitely many Taylor coefficients of (4.2) occur. In particular the finite anomaly is an explicit polynomial in Euler’s constant and ordinary zeta values, with rational functions of the rates and logarithms of their positive differences as coefficients. This is a local conclusion; it does not evaluate the merged global constant Q_d .

For equally spaced triples, (4.13) recovers the earlier three-factor contact specialization [6]:

$$\text{CT}_{\epsilon \downarrow 0} C_3(0, \epsilon, 2\epsilon) = Q_3 + \frac{\zeta(2)}{2} \log 2.$$

For four equally spaced points, the new formula is

$$\text{CT}_{\epsilon \downarrow 0} C_4(0, \epsilon, 2\epsilon, 3\epsilon) = Q_4 + \gamma \zeta(2)(2 \log 2 + \log 3) - \frac{3}{2} \zeta(3)(3 \log 2 + \log 3). \quad (4.14)$$

4.3 A hierarchical collision

Take shifts $(0, \epsilon, \epsilon + \epsilon^2)$, so the last pair approaches one another on a smaller scale than its distance from the first point. The four terms in (4.4) are exactly

$$\begin{aligned} T_{\{1,2,3\}} &= \frac{\log \epsilon}{\epsilon^3} - \frac{\log(\epsilon + \epsilon^2)}{\epsilon^3(1 + \epsilon)}, \\ T_{\{1,2\}} &= \frac{h(\epsilon^2)}{\epsilon} \log \epsilon, \\ T_{\{1,3\}} &= \frac{h(-\epsilon^2)}{\epsilon + \epsilon^2} \log(\epsilon + \epsilon^2), \\ T_{\{2,3\}} &= \frac{h(-\epsilon - \epsilon^2)}{\epsilon^2} \log(\epsilon^2). \end{aligned}$$

Writing $\Lambda = \log \epsilon$, expanding these exact terms, and applying Theorem 4.1 yields

$$\begin{aligned} C_3(0, \epsilon, \epsilon + \epsilon^2) &= \frac{(1 + 2\gamma)\Lambda - 1}{\epsilon^2} + \frac{(2\gamma + 2\zeta(2) - 1)\Lambda + \frac{3}{2}}{\epsilon} \\ &\quad + (1 - \gamma + 2\zeta(2) + 2\zeta(3))\Lambda + Q_3 + \gamma - \frac{11}{6} + O(\epsilon|\log \epsilon|). \end{aligned} \quad (4.15)$$

The finite constant is therefore $Q_3 + \gamma - 11/6$. The rational term $-11/6$ comes from the all-pole subset, while the term γ comes from $T_{\{1,3\}}$. This example shows why one cannot obtain hierarchical constant terms by retaining only the singular powers of a fixed-rate expansion and subsequently varying its rate parameters.

4.4 Verification and precise scope

The companion script `verify_collision.py` verifies the quartic constant and the full hierarchical expansion through order zero by exact symbolic algebra. It also evaluates the separated circle integral by subtracting its actual right-hand poles interval by interval, independently of (4.4). Direct Laurent subtraction at the merged endpoint gives

$$\begin{aligned} Q_3 &= -1.9662627343915343406643753384116\dots, \\ Q_4 &= 13.510120376015827411606240712830\dots \end{aligned}$$

The computed differences $C_d - T_d - Q_d$ tend to zero linearly on both displayed paths, at four values of ϵ for each path and 55 working decimal digits. The separate script `verify_derivative_pair.py` checks both displayed derivative counterterms at the same four gaps, using independent double-pole subtraction and 65 working decimal digits. Its four comparisons with (4.12) have observed absolute residual at most 6.4×10^{-51} . The package includes the raw JSON records. These floating-point diagnostics check implementation and signs; the interpolation proof supplies the theorem.

The result settles the arbitrary-factor and arbitrary argument-derivative parts of the unit-frequency geometric collision question at Stieltjes index zero, including hierarchical approaches. Higher Stieltjes indices introduce logarithmic singular germs and are not covered by the rational interpolation argument. Associativity of successively renormalized partial mergers also remains a separate question: an exact simultaneous subtraction does not by itself identify every iterated subtraction rule. Two further extensions are a complete statement for integer-frequency singular grids, with each local pole amplitude scaled in the ambient coordinate, and a weighted version for analytic periodic test functions which would provide collision formulas for Fourier moments. Neither is included in the theorems proved here.

5 The complete quartic directional layer of Tornheim zeta

This section answers the explicit quartic-layer question in the incoming *Cubic Tornheim Derivatives and Harmonic Stieltjes Constants*, Section 7, Question 4 [3], and the corresponding question in *Independent Orders and Exact Reductions*, Section 7 [2]. The incoming work already proves the cubic ray formula and evaluates its diagonal coordinate in a convergent harmonic series. Those results are not new here. The present extension classifies the next homogeneous layer, gives convergent representatives for every coordinate, and obtains exact quartic cancellation identities.

Put

$$L = \log(2\pi), \quad d_j = \zeta^{(j)}(0), \quad e_j = \zeta^{(j)}(-1), \quad Z_j = \zeta(j), \quad g = \gamma.$$

Spectral derivatives are denoted by parentheses on zeta, whereas subscripts on a multivariable analytic function mean partial derivatives. The classical Mellin continuation [10] of

$$T(A, B, C) = \sum_{m, n \geq 1} m^{-A} n^{-B} (m + n)^{-C}$$

has the following local form, already established in the incoming independent-order report [2]:

$$T(A, B, C) = \zeta(A)\zeta(B) - \frac{C\zeta(B-1)}{A+C} - \frac{C\zeta(A-1)}{B+C} + CR(A, B, C), \quad (5.1)$$

where R is holomorphic near the origin and symmetric in A, B . The two indicated polar quotients must be restricted before evaluating at the origin. Thus for $(a+c)(b+c) \neq 0$ the function $W_{a,b,c}(t) = T(at, bt, ct)$ is holomorphic near zero, although T is not jointly holomorphic there.

5.1 Six cubic coefficients and three exact constraints

All derivatives in the next display are evaluated at $(0, 0, 0)$:

$$\alpha = R_{AAA}, \quad \beta = R_{AAB}, \quad \delta = R_{AAC}, \quad \varepsilon = R_{ABC}, \quad \varphi = R_{ACC}, \quad \chi = R_{CCC}.$$

Symmetry gives the corresponding derivatives with A, B interchanged. These six quantities describe the cubic homogeneous part of R .

Proposition 5.1. *One has*

$$\chi = \frac{e_4 - d_4}{4}, \quad \delta = \frac{d_2^2 - d_4}{4}, \quad \alpha + 3\varphi = -\frac{Ld_3}{2} - d_4. \quad (5.2)$$

If $\Omega_4 = W_{1,1,1}^{(4)}(0)$, then

$$\boxed{\Omega_4 = 24(\beta + \varepsilon) - 8Ld_3 + 12d_2^2 - 16d_4.} \quad (5.3)$$

Proof. The exact axis $T(0, 0, C) = \zeta(C-1) - \zeta(C)$ gives

$$CR(0, 0, C) = \zeta(C-1) - \zeta(C) - \frac{5}{12}.$$

Its coefficient of C^4 proves the first assertion. The Euler stuffle identity on the plane $B = 0$ gives

$$\begin{aligned} CR(A, 0, C) + AR(C, 0, A) &= \zeta(A)\zeta(C) - \zeta(A+C) + \frac{1}{2}(\zeta(A) + \zeta(C)) \\ &\quad + \zeta(-1) + \zeta(A-1) + \zeta(C-1). \end{aligned} \quad (5.4)$$

The coefficient of A^2C^2 is δ on the left and $(d_2^2 - d_4)/4$ on the right. The coefficient of A^3C is $\alpha/6 + \varphi/2$ on the left and $d_1d_3/6 - d_4/6$ on the right. Since $d_1 = -L/2$, this proves the remaining two constraints.

On the diagonal, (5.1) becomes $W_{1,1,1}(t) = \zeta(t)^2 - \zeta(t-1) + tR(t, t, t)$. Its fourth derivative is

$$-d_4 - 4Ld_3 + 6d_2^2 - e_4 + 8\alpha + 24\beta + 24\delta + 24\varepsilon + 24\varphi + 4\chi.$$

Substituting (5.2) proves (5.3). □

The three constraints have rank three in the six-dimensional space of formal cubic jets symmetric in A, B . Thus that formal reconstruction problem has exactly three free coordinates. This dimension statement does not imply arithmetic independence of the evaluated constants.

5.2 An explicit formula for every fourth ray derivative

Theorem 5.2. For $a, b, c \in \mathbb{C}$ with $(a + c)(b + c) \neq 0$,

$$\begin{aligned}
 W_{a,b,c}^{(4)}(0) = & abc^2\Omega_4 + 4c(a+b)(a^2 - ab + b^2 - c^2)\alpha \\
 & + 12abc(a+b-2c)\beta \\
 & - 2[ab(a^2 + b^2) - 4abc^2 + c^3(a+b)]Ld_3 \\
 & + 3[2a^2b^2 + c^2(a^2 + b^2 - 4ab)]d_2^2 \\
 & + [-\tfrac{1}{2}(a^4 + b^4) - 3c^2(a^2 + b^2) + 16abc^2 - 4c^3(a+b) - c^4]d_4 \\
 & + \left[c^4 - \frac{cb^4}{a+c} - \frac{ca^4}{b+c} \right] e_4.
 \end{aligned} \tag{5.5}$$

The coordinates α, β, Ω_4 have the convergent Gamma and polylogarithmic representations below; no unspecified partial derivative is required to evaluate the formula.

Proof. Expanding the holomorphic restriction of (5.1) gives

$$\begin{aligned}
 W_{a,b,c}^{(4)}(0) = & -\tfrac{1}{2}(a^4 + b^4)d_4 - 2ab(a^2 + b^2)Ld_3 + 6a^2b^2d_2^2 \\
 & - c \left(\frac{b^4}{a+c} + \frac{a^4}{b+c} \right) e_4 + 4c\alpha(a^3 + b^3) + 12c\beta ab(a+b) \\
 & + 12c^2\delta(a^2 + b^2) + 24abc^2\varepsilon + 12c^3\varphi(a+b) + 4c^4\chi.
 \end{aligned}$$

Eliminate δ, φ, χ using (5.2), then eliminate ε with (5.3). Factoring the α coefficient proves the claimed formula. $\square$

For instance,

$$W_{1,1,2}^{(4)}(0) = 4\Omega_4 - 48\alpha - 48\beta - 4Ld_3 - 18d_2^2 - 41d_4 + \tfrac{44}{3}e_4, \tag{5.6}$$

$$W_{1,2,2}^{(4)}(0) = 8\Omega_4 - 24\alpha - 48\beta - 4Ld_3 - 12d_2^2 - \tfrac{105}{2}d_4 + \tfrac{29}{6}e_4. \tag{5.7}$$

The three positive anchor rays $(1, 1, 1), (1, 1, 2), (1, 2, 2)$ recover all three formal coordinates: the determinant of their coefficient matrix in $(\Omega_4, \alpha, \beta)$ is 1152.

5.3 Convergent representatives: a Gamma moment and two kernels

The incoming boundary-jet hierarchy [2] already gives, with

$$\rho(n) = \log \Gamma(n+1) - (n + \tfrac{1}{2}) \log n + n - \frac{L}{2} - \frac{1}{12n},$$

the identity

$$\alpha = - \sum_{n \geq 1} \rho(n) \log^3 n - e_4 - \tfrac{1}{2}d_4 - e_3 - \tfrac{\gamma_3}{12}. \tag{5.8}$$

The series converges absolutely because $\rho(n) = O(n^{-3})$. Its use as a quartic coordinate is new here; the all-order boundary identity itself is credited to the incoming report.

For the other two coordinates, define for $x > 0$

$$f_j(x) = \partial_s^j \text{Li}_s(e^{-x}) \Big|_{s=0}, \quad p_1(x) = \frac{g + \log x}{x}, \quad p_2(x) = \frac{(g + \log x)^2 + Z_2}{x}.$$

Put

$$\begin{aligned} S_{21}(x) &= p_2(x)p_1(x) + p_2(x)d_1 + p_1(x)d_2 - x(p_2(x)e_1 + p_1(x)e_2) + d_2d_1, \\ S_{11}(x) &= p_1(x)^2 + 2p_1(x)d_1 - 2xp_1(x)e_1 + d_1^2. \end{aligned}$$

The ordinary convergent integrals are

$$J_{21} = \int_0^1 \frac{f_2(x)f_1(x) - S_{21}(x)}{x} dx + \int_1^\infty \frac{f_2(x)f_1(x)}{x} dx, \quad (5.9)$$

$$K_{11} = \int_0^1 \frac{\log x + g}{x} (f_1(x)^2 - S_{11}(x)) dx + \int_1^\infty \frac{\log x + g}{x} f_1(x)^2 dx. \quad (5.10)$$

Set $\mu_2 = g^2 + Z_2$ and $\mu_3 = g^3 + 3gZ_2 + 2Z_3$, and define

$$\begin{aligned} E_{21} &= gd_1d_2 - (\mu_2 + 2g + 2)d_1 - (g + 1)d_2 - \frac{\mu_3e_1}{3} - \frac{\mu_2e_2}{2} \\ &\quad - \frac{3}{8} - \frac{3g}{4} - \frac{\mu_2}{4} - \frac{g^2}{2} - \frac{g\mu_2}{2}, \end{aligned} \quad (5.11)$$

$$\begin{aligned} E_{111} &= -\frac{g^3}{2} - \frac{3g^2}{4} - \frac{3g}{4} - \frac{3}{8} - 2(g^2 + 2g + 2)d_1 \\ &\quad + \frac{g^2 - Z_2}{2}d_1^2 - \frac{2(g^3 - Z_3)}{3}e_1. \end{aligned} \quad (5.12)$$

Proposition 5.3. *The integral and constant formulas give*

$$\boxed{\beta = J_{21} + E_{21}, \quad \varepsilon = K_{11} + E_{111}.} \quad (5.13)$$

Together with (5.3), these evaluate Ω_4 using two ordinary, absolutely convergent one-dimensional integrals.

Proof. For parameters near zero put $G(C) = 1/\Gamma(1 + C)$ and

$$\begin{aligned} S(A, B; x) &= \Gamma(1 - A)\Gamma(1 - B)x^{A+B-2} + \Gamma(1 - A)\zeta(B)x^{A-1} + \Gamma(1 - B)\zeta(A)x^{B-1} \\ &\quad - \Gamma(1 - A)\zeta(B - 1)x^A - \Gamma(1 - B)\zeta(A - 1)x^B + \zeta(A)\zeta(B), \\ I(A, B, C) &= \int_0^1 x^{C-1} (\text{Li}_A(e^{-x}) \text{Li}_B(e^{-x}) - S(A, B; x)) dx \\ &\quad + \int_1^\infty x^{C-1} \text{Li}_A(e^{-x}) \text{Li}_B(e^{-x}) dx, \\ X(A, B, C) &= \frac{\Gamma(1 - A)\Gamma(1 - B)}{A + B + C - 2} + \frac{\Gamma(1 - A)\zeta(B)}{A + C - 1} + \frac{\Gamma(1 - B)\zeta(A)}{B + C - 1}. \end{aligned}$$

Jonquière's expansion at zero [15, Eq. 25.12.12] shows that the subtracted integrand and each fixed parameter derivative are dominated by an integrable power times a polynomial in $|\log x|$ on a sufficiently small common polydisc. The tail is exponentially decreasing. Thus I is holomorphic. Integrating the six subtractions explicitly in the classical Mellin representation, then using (5.1), gives

$$\begin{aligned} R(A, B, C) &= G(C)(I(A, B, C) + X(A, B, C)) + \frac{G(C) - 1}{C} \zeta(A)\zeta(B) \\ &\quad - \frac{\Gamma(1 - A)G(C) - 1}{A + C} \zeta(B - 1) - \frac{\Gamma(1 - B)G(C) - 1}{B + C} \zeta(A - 1). \end{aligned} \quad (5.14)$$

Every displayed quotient is removable at the origin. For example, $\Gamma(1 - A)G(-A) = 1$ proves divisibility by $A + C$.

The derivative $I_{AAB}(0)$ is exactly J_{21} , since differentiating the six subtractions gives S_{21} . Likewise $I_{ABC}(0) + gI_{AB}(0) = K_{11}$, since $G'(0) = g$ and the mixed subtraction is S_{11} . The remaining derivatives in (5.14) are finite Gamma and zeta algebra. In particular,

$$\partial_A \partial_C \frac{\Gamma(1 - A)G(C) - 1}{A + C} \Big|_0 = \frac{g^3 - Z_3}{3}.$$

For $C = 0$, the derivatives of $(\Gamma(1 - A) - 1)/A$ of orders 1, 2 are $\mu_2/2, \mu_3/3$. These evaluations and the elementary denominators in X give exactly E_{21} and E_{111} .

Finally $f_2 f_1 - S_{21} = O(x(1 + |\log x|^2))$ and $f_1^2 - S_{11} = O(x(1 + |\log x|))$ near zero. Their displayed integrals therefore converge absolutely in the ordinary sense. $\square$

5.4 Exact cancellation families

Corollary 5.4 (Cyclic quartic reduction). *Suppose all pairwise sums of a, b, c are nonzero, and let $s_1 = a + b + c$, $s_2 = ab + bc + ca$, $s_3 = abc$. Then*

$$\begin{aligned} \sum_{\text{cyc}} W_{a,b,c}^{(4)}(0) &= s_1 s_3 \Omega_4 + (-4s_1^2 s_2 + 8s_2^2 + 12s_1 s_3) Ld_3 \\ &\quad + (12s_2^2 - 36s_1 s_3) d_2^2 \\ &\quad + (-2s_1^4 + 4s_1^2 s_2 - 2s_2^2 + 24s_1 s_3) d_4. \end{aligned} \quad (5.15)$$

Both off-diagonal coordinates and e_4 cancel identically.

Proof. The α coefficient is $4 \sum_{\text{cyc}} c(a^3 + b^3) - 4 \sum_{\text{cyc}} c^3(a + b) = 0$. The β coefficient is $12abc \sum_{\text{cyc}} (a + b - 2c) = 0$. For the e_4 coefficient, the terms with denominator $a + c$ combine to $-b^4$ and cancel the sum of fourth powers. The remaining symmetric polynomials yield (5.15). $\square$

There is also a useful one-parameter source of completely evaluated differences. On the additive plane $c = a + b$, the nonclassical part of (5.5) is

$$ab(a + b)^2(\Omega_4 - 12\alpha - 12\beta).$$

Any two admissible rays on this plane therefore have a weighted difference involving only ordinary zeta derivatives. A small example is

$$\boxed{W_{1,2,3}^{(4)}(0) - \frac{9}{2} W_{1,1,2}^{(4)}(0) = -20Ld_3 + 24d_2^2 - 76d_4 + \frac{12}{5}e_4.} \quad (5.16)$$

This is an evaluated identity with no undetermined Tornheim coordinate.

For a single ray with $abc \neq 0$, the two displayed off-diagonal coefficients vanish simultaneously if and only if $a = b = c$. Indeed the β coefficient forces $a + b = 2c$; substituting this into the α coefficient gives a nonzero multiple of $(a - b)^2$. Thus the nontrivial cubic cancellation conic from the incoming report does not persist as a conic at the fourth layer. This assertion concerns the stated formal-coordinate elimination. An additional arithmetic relation among the constants could enlarge the family of values reducible in a chosen constant algebra.

5.5 All-order jet structure and the limit of cyclic diagonal reduction

The quartic formula is the last layer at which the axis and Euler stuffle constraints force a cyclic sum to depend on only one diagonal coordinate. The following exact polynomial statement explains why.

Fix an integer $n \geq 3$. Subtract two possible degree- n homogeneous jets of (5.1) whose axis and Euler boundary restrictions agree. Their difference is a homogeneous polynomial $U(A, B, C)$ of degree n satisfying

$$U(A, B, C) = U(B, A, C), \quad C \mid U, \quad U(A, 0, C) + U(C, 0, A) = 0. \quad (5.17)$$

These are conditions on a formal reconstruction space, not additional functional equations asserted for arbitrary perturbations of T .

Theorem 5.5 (Normal form of the residual jet). *Every polynomial satisfying (5.17) has a unique representation*

$$U = C[A(A - C)V(A, C) + B(B - C)V(B, C)] + ABCQ(A, B, C), \quad (5.18)$$

where V is a symmetric two-variable homogeneous polynomial of degree $n - 3$, and Q is homogeneous of degree $n - 3$ and symmetric in A, B . Consequently this residual space has dimension

$$D_n = \left\lfloor \frac{n^2 - 1}{4} \right\rfloor. \quad (5.19)$$

For $n = 1, 2$ the corresponding residual space is zero.

Proof. The boundary polynomial $F(A, C) = U(A, 0, C)$ is antisymmetric in A, C . It is divisible by C , hence by A as well, and antisymmetry also implies divisibility by $A - C$. Therefore $F(A, C) = AC(A - C)V(A, C)$, where V is symmetric and has degree $n - 3$. This uniquely determines V . Subtract the first term of (5.18) from U . The difference vanishes on both $A = 0$ and $B = 0$, by symmetry, and remains divisible by C . Hence it is uniquely $ABCQ$, with the required symmetry. Conversely the displayed normal form satisfies all three conditions.

The symmetric two-variable polynomials of degree $n - 3$ have dimension $\lfloor (n - 1)/2 \rfloor$. Polynomials in three variables of that degree which are symmetric in the first two have dimension $\lfloor (n - 1)^2/4 \rfloor$, by counting the exponent pairs of A, B . Their sum is (5.19). If $n < 3$, the two divisibility arguments force $U = 0$ directly. $\square$

Corollary 5.6 (Cyclic residual dimension). *The cyclic symmetrization of the residual space is precisely*

$$\{ABC S(A, B, C) : S \text{ is symmetric in all three variables and homogeneous of degree } n - 3\}.$$

Its dimension is

$$p_{\leq 3}(n - 3) = \left\lfloor \frac{n^2 + 3}{12} \right\rfloor, \quad (5.20)$$

where $p_{\leq 3}(m)$ is the number of partitions of m into at most three parts. In particular this dimension is one at orders three and four, and is two at order five.

Proof. The terms involving V in (5.18) cancel in pairs under cyclic summation, because $V(A, C) = V(C, A)$. The remaining sum is $ABC \sum_{\text{cyc}} Q(A, B, C)$. Since Q is symmetric in its first two variables, this sum is fully symmetric. Conversely any symmetric S occurs by taking $Q = S/3$ and $V = 0$.

Symmetric homogeneous polynomials of degree m have basis $s_1^i s_2^j s_3^k$ with $i + 2j + 3k = m$. Counting these triples, or expanding $(1-t)^{-1}(1-t^2)^{-1}(1-t^3)^{-1}$, gives $p_{\leq 3}(m) = \lfloor (m^2 + 6m + 12)/12 \rfloor$, proving (5.20). $\square$

Evaluation on the diagonal is a nonzero linear functional on this cyclic residual space. It therefore determines the whole space when the dimension is one, explaining the complete cyclic formulas at orders three and four. The following explicit fifth-order polynomial shows what fails next:

$$U_5(A, B, C) = ABC[(A + B + C)^2 - 3(AB + BC + CA)]. \quad (5.21)$$

It satisfies the boundary conditions, vanishes on the entire diagonal, and has nonzero cyclic sum $3U_5$. Consequently the axis, symmetry, Euler stuffle, and diagonal data alone cannot force a fifth-order cyclic formula involving just the diagonal fifth derivative and ordinary zeta jets. This is a precise limitation of those input identities; other functional equations may provide additional constraints.

5.6 Verification and remaining arithmetic questions

The companion implementation `check_quartic.py` checks the full rational-function identity, cyclic elimination, diagonal and Euler-diagonal cases, and both elementary integral correction terms. It evaluates α with a finite Stirling expansion and computes J_{21}, K_{11} by an integrated Jonquière series on $(0, 1)$ plus exponentially convergent series on $(1, \infty)$. A separate evaluator computes the fourth ray derivative by a fifteen-node symmetric difference of the original Mellin continuation. These routes give

$$\begin{aligned} \alpha &= 13.1294216474884019236329146797203 \dots, \\ \beta &= 6.90381599352243710775587231649323 \dots, \\ \varepsilon &= 4.68913245627221585695112599939183 \dots, \\ \Omega_4 &= 798.777376093189588968747119645501 \dots \end{aligned}$$

At 55 decimal digits of working precision, four independently evaluated rays $(1, 1, 1)$, $(1, 0, 2)$, $(1, 2, 2)$, and $(1, -2, 3)$ had absolute discrepancies below $8.2 \cdot 10^{-34}$. These decimal checks use `mpmath` and are not interval certificates; the exact identities rest on the proofs above.

The result completes the quartic finite-coordinate problem posed in the incoming reports. It does not provide a finite expression for Ω_4, α, β in a smaller algebra of ordinary constants. Natural next questions are: relate the two new integral coordinates to higher harmonic Stieltjes coefficients; identify the dimension and canonical anchor directions at every order; obtain general cyclic elimination laws; and formalize the finite jet algebra separately from the Mellin continuation and normal-convergence lemmas.

6 Audit, interpretation, and further research

6.1 What has been proved, and what has not

The following table records the mathematical status at the inspected revision. The word “resolved” refers to the explicitly stated component of a research question, not to every possible arithmetic refinement.

Question or object	Result and scope
Nonpolynomial reflected generator	Theorems 2.1 and 2.2 give a complex-power sine family, its polylogarithmic Euler representation, and the full joint completion. All Stieltjes indices, all argument-derivative orders, and all logarithmic powers are covered.
First nonlinear log-sine bridge	Theorem 2.7 expresses the sine-square moment through the inclusive harmonic Stieltjes coefficient and then eliminates that coefficient in (2.30). The separate cubic bridge is inherited from [3].
Centered harmonic cancellation	Theorem 3.1 is necessary and sufficient for every polynomial in finitely many generalized harmonic variables. It is stronger than checking finitely many residues. Theorem 3.6 evaluates an infinite family of its regular values.
Geometric collisions	Theorems 4.1 and 4.3 cover arbitrary finite clusters and argument-derivative orders at Stieltjes index zero and unit frequency, with no comparability requirement on the gaps. Higher Stieltjes indices and iterated merger associativity remain open here.
Quartic directional Tornheim jets	Theorem 5.2 reconstructs every admissible fourth ray derivative from three specified coordinates. Proposition 5.3 gives convergent representatives. The difference (5.16) eliminates all three coordinates.
Higher formal relation spaces	Theorem 5.5 and Corollary 5.6 give exact dimensions for the stated axis, symmetry, and Euler boundary system. They do not prove period independence or completeness of all functional identities.
Gaussian S_6 and revised S_8	Still conjectural. The present work supplies neither a period certificate nor a counterexample. Existing formal nonmembership results restrict their specified search systems only [1].

The phrase “finite-coordinate formula” is deliberately separate from “evaluation in ordinary constants.” For example, the quartic theorem is effective: its retained coordinates are given by convergent Gamma series and polylogarithmic integrals. A further reduction of those coordinates is an additional arithmetic theorem. Conversely, (2.30) and (5.16) already are evaluations in the displayed ordinary constants, because the residual coordinates cancel algebraically.

6.2 Corrections and qualifications to preserve at integration

No additional false theorem was established in the focused portions of the nine incoming reports that were checked. Several previously documented corrections should retain their original attribution: the rejected earlier S_8 coefficient vector, the restricted sector required in the Crandall–Jonquière expansion, the extra harmonic factor in an external boundary formula, and the distinction between a numerical period conjecture and formal row-space nonmembership [1, 3, 2]. They are not discoveries of the present article.

The new results do rule out three plausible but unjustified extensions. They suggest the following concrete wording for future integration.

1. **Specify the entire crossing numerator.** Replace any convention saying only “take the removable value of the regularized integral” by the holomorphic completion (2.10), or its

locally equivalent full subtraction. At $p = 2$, the uncompleted analytic limit is $2\zeta(2)$ while the coordinate finite part is zero. Merely observing that the residue vanishes at the intersection does not remove its finite contribution.

2. **Retain boundary contacts under argument differentiation.** The two derivative correlations satisfy $C_{(1,0)} + C_{(0,1)} = \pi^2 \csc^2(\pi a)$, not zero, in the specified finite-part convention. A formal integration-by-parts or translation argument that drops the seam terms loses this exact contribution. Use the direct repeated-pole calculation or an explicitly compatible distributional extension.
3. **Limit the diagonal cyclic claim to its proven orders.** The polynomial U_5 in (5.21) is invisible to the axis, boundary, and diagonal data but has a nonzero cyclic sum. Thus the cubic and quartic reduction to one diagonal coordinate cannot be extended to all orders using those relations alone. Additional relations must be stated and proved.

There is a related algebraic warning in Section 3. Vanishing at a finite collection of spectral centers is not evidence for the all-center property. The explicit near-miss polynomials following (3.8) show exactly where this inference fails. The replacement is the finite polynomial identity $\iota(P) = P$, whose necessity uses the proved formal independence lemma.

6.3 Reproducibility and evidence

Each theorem has an analytic or exact algebraic proof in the article. The Python files independently check sensitive signs, finite constants, and symbolic eliminations. They require only Python, SymPy, and mpmath; no source archive or network connection is needed to run them. The verification README explains the methods, recorded versions, precision, and report-writing behavior.

Artifact	Independent check
<code>check_beta.py</code>	Direct Hurwitz-weight quadrature versus finite Fourier sums and the Euler–polylog integral; Gamma reflection moments; separately subtracted digamma logarithmic and cubic integrals; exact Fourier jets.
<code>centered_checks.py</code>	Exact Bernoulli recurrences, affine derivative relations, parity examples, first-pole counterexamples, and Faulhaber coefficients; independent finite-head and Hurwitz-tail values for $M = 0, \dots, 6$.
<code>check_quartic.py</code>	Exact rational reconstruction, cyclic cancellation, integral correction constants, dimensions, and the fifth-order obstruction; four fourth derivatives evaluated from the original continued Mellin expression.
<code>verify_collision.py</code>	Exact quartic and hierarchical coefficient extraction; direct interval-by-interval finite parts compared with the predicted merged limit after subtraction.
<code>verify_derivative_pair.py</code>	Independent double-pole quadrature for the two derivative counterterms and the exact trigamma-reflection sum.

The beta identities, except for the integrable singular Euler test, have observed numerical discrepancies below $1.3 \cdot 10^{-60}$ in the recorded run; the Euler test has discrepancy $2.5 \cdot 10^{-33}$ or less. The four Tornheim ray comparisons are below $8.2 \cdot 10^{-34}$. The derivative-pair reflection checks

are below $6.4 \cdot 10^{-51}$. These are floating-point observations at the documented precisions. For geometric collisions, $C - T - Q$ at a finite separation is a remainder that should tend to zero, not an equality error expected to be tiny at each separation. The centered checks have an additional explicit analytic truncation bound proved in the appendix; rounding errors have not been enclosed by interval arithmetic.

6.4 Proposed research questions

The following questions are intended as concrete next projects. They emphasize exact identities and structure; the analytic estimates needed to justify an interchange or a numerical reproduction are auxiliary.

Research question 6.1 (Higher log-sine cancellations). Determine the smallest harmonic-zeta Laurent family needed for $J_k(s)$ at each $k \geq 3$, using the explicit coefficient formula (2.20). Which linear combinations of finite-part products of digamma, polygamma, and logarithmic sine powers cancel every nonordinary harmonic coordinate? The first target is the next layer after (2.30), with its relation matrix derived symbolically before any numerical search.

The generator fixes both inclusive harmonic conventions and all endpoint constants. This makes it possible to compare candidate combinations without changing regularization halfway through the calculation.

Research question 6.2 (An arithmetic description of quartic coordinates). Express J_{21} and K_{11} , or their canonical linear combinations, through Laurent coefficients of explicitly defined harmonic multiple zeta functions. Does the combination $\beta + \varepsilon$ admit a shorter one-dimensional harmonic series analogous to the cubic bridge? Can any of Ω_4, α, β be reduced further in a specified ordinary constant algebra?

The convergent representations here provide exact starting expressions. A numerical integer relation can guide a target but would not resolve the question.

Research question 6.3 (The second quintic cyclic coordinate). Compute the full fifth-order cyclic formula. After the known boundary part is removed, its residual lies in $ABC \operatorname{span}\{(A+B+C)^2, AB+BC+CA\}$. The diagonal value and one suitable nondiagonal cyclic value, for example at $(1, 1, 2)$, separate these two coordinates. Find convergent representatives and short combinations in which both cancel.

The assertion about the two evaluations is finite algebra: the ratios s_1^2/s_2 are 3 at $(1, 1, 1)$ and $16/5$ at $(1, 1, 2)$. What remains is to compute the ordinary part and the new analytic coordinate, not to conjecture the dimension.

Research question 6.4 (Canonical ray sets at arbitrary order). Use the normal form (5.18) to choose small positive integer anchor rays giving an invertible evaluation matrix of size D_n . For cyclic reconstruction the corresponding size is $p_{\leq 3}(n-3)$. Can the inverses and the resulting cancellation identities be organized uniformly in n ?

One should also test additional global Tornheim functional equations against these spaces. They may reduce the dimension of the actual reconstruction problem; the dimensions in this article are exact only for their stated input relations.

Research question 6.5 (Higher Stieltjes geometric collisions). Replace $\gamma_0^{(r_i)}$ by $\gamma_{m_i}^{(r_i)}$ in Theorem 4.3. The singular local germs now contain powers of $\log x$. Construct a simultaneous Mellin-parameter completion whose derivatives give explicit counterterms uniformly for hierarchical gaps, and prove that the remainder extends through the total collision.

The existing all-factor spectral completions are relevant input, but they do not by themselves justify uniform passage through vanishing geometric gaps. This is the missing analytic statement to isolate.

Research question 6.6 (Integer-frequency grids and Fourier weights). Give the complete geometric theorem for $\prod_i \gamma_0^{(r_i)}(\{p_i x + s_i\})$, with positive integers p_i , optionally multiplied by an analytic periodic weight. At each merging grid point the local shifts are s_i/p_i and the pole amplitudes must be scaled in the ambient coordinate. Derive the resulting finite Fourier-moment contact formulas and their conductor dependence.

The local interpolation proof strongly suggests this extension. A complete statement must inventory all grid intersections and preserve their coordinate scale; it is not included in the proved theorems here.

Research question 6.7 (Compatibility of successive mergers). Characterize subtraction rules for partial clusters whose iterated mergers agree with the simultaneous full subtraction. For three and four factors, calculate the finite difference between two merger orders and decide whether a universal local adjustment makes the operation associative.

The hierarchical example (4.15) supplies an exact test case. Existence of the fully subtracted limit does not settle associativity of other subtraction prescriptions.

Research question 6.8 (Centered products beyond a trigamma square). Evaluate all centered even moments of products of an even number of even-order harmonic tails. Start with $(H_n^{(2r)} - \zeta(2r))(H_n^{(2q)} - \zeta(2q))$, then four tails. Determine which families reduce by finite summation by parts to ordinary Euler sums and which require higher-depth constants.

The pole cancellation is already completely answered by Theorem 3.1; the new task is evaluation of the regular values. This separates a solved analytic classification from an arithmetic problem that can grow with the number of factors.

Research question 6.9 (Other outer shifts and periodic harmonic data). Classify the polynomials P for which $\sum_{n \geq 0} P(H_n^{(1)}, \dots, H_n^{(R)})(n+a)^{-s}$ is regular at all nonpositive even integers when $a \neq 1/2$. Then consider harmonic variables with rational inner shifts or periodic coefficients. What replaces the reflection involution, and how do coincident translation-pole orbits affect the independence argument?

The additive-difference lemma reduces this to a concrete analysis of rational telescoping in each pole orbit. It should be proved for the specific family rather than inferred from a few vanished Bernoulli coefficients.

Research question 6.10 (A joint beta-collision generator). Combine the complex-power sine weight with moving periodic Stieltjes factors. Determine the polar arrangement in the weight, spectral, and gap parameters, and give a canonical subtraction whose restrictions recover both (2.10) and (4.11).

Such a formula would connect Fourier coefficients, Gamma primitives, and geometric contact terms in a single analytic object. It would also identify explicitly which order of restriction is compatible with the unit-coordinate finite part.

Research question 6.11 (Return to the Gaussian period conjectures). Find a functional identity or period certificate for the frozen S_6 and revised S_8 candidates outside the already obstructed relation system. Can a harmonic or Stieltjes transform supply a new exact relation after a justified change of variables to the Gaussian alphabet?

No such bridge is proved here. The source vectors and their conventions should remain fixed, and a proposed proof must produce an exact certificate for those vectors rather than a nearby fitted identity.

Research question 6.12 (Formalization in ProveIt). Separate the formal development into reusable analytic and algebraic lemmas: finite asymptotic subtraction with normal convergence; coefficient extraction from a completed meromorphic germ; confluent interpolation on a fixed contour; the additive-difference lemma; and the homogeneous residual-space normal form. Which of the present finite identities can then be generated and checked automatically from these lemmas?

The supplied exact computations can serve as regression examples. They are not proof-assistant certificates, and incorporation into the repository should not silently confer a formalized status.

A Analytic control of the centered verification tail

The numerical checks of the centered trigamma-square moments use a finite head and an explicitly summed Bernoulli tail. The tail truncation admits the following analytic bound. This bound controls truncation of an asymptotic expansion; the recorded `mpmath` evaluations use ordinary floating-point arithmetic and are not claimed to be interval enclosures. The mathematical identity is proved separately in Theorem 3.6.

For real $x > 0$, write

$$T(x) = \psi'(x + 1/2), \quad A_K(x) = \sum_{j=0}^K B_{2j}(1/2)x^{-2j-1}, \quad C_K = \frac{2(2K+2)!}{(2\pi)^{2K+2}}.$$

For every integer $K \geq 0$,

$$0 < T(x) \leq x^{-1}, \quad |T(x) - A_K(x)| \leq C_K x^{-2K-3}. \quad (\text{A.1})$$

Indeed the standard trigamma integral gives

$$T(x) = \int_0^\infty e^{-xt} g(t) dt, \quad g(t) = \frac{t}{2 \sinh(t/2)}.$$

The partial fraction expansion of the hyperbolic cosecant [16, Eq. 4.36.5] is

$$g(t) = 1 + 2 \sum_{k=1}^\infty (-1)^k \frac{t^2}{t^2 + (2\pi k)^2}.$$

Expand each rational term through degree $2K$. With $a_k = 2\pi k$, the remainder is exactly

$$2(-1)^K t^{2K+2} \sum_{k=1}^\infty \frac{(-1)^k}{a_k^{2K} (t^2 + a_k^2)}.$$

The positive summand magnitudes decrease in k , so the alternating series estimate yields

$$\left| g(t) - \sum_{j=0}^K \frac{B_{2j}(1/2)}{(2j)!} t^{2j} \right| \leq \frac{2t^{2K+2}}{(2\pi)^{2K+2}}.$$

The finite polynomial is identified by the Bernoulli generating function $g(t) = te^{t/2}/(e^t - 1)$. Integration proves the second inequality in (A.1); the first follows from $0 < g(t) \leq 1$.

Fix integers $M \geq 0$, $K \geq M$, and $N \geq 1$, and set $a = N + 1/2$. Let

$$b_j = B_{2j}(1/2), \quad d_\ell^{(K)} = \sum_{\substack{0 \leq j \leq K \\ 0 \leq \ell - j \leq K}} b_j b_{\ell-j}, \quad C_A = \sum_{j=0}^K |b_j| a^{-2j}.$$

The nondecaying polynomial to subtract from $x^{2M}T(x)^2$ is

$$P_M(x) = \sum_{\ell=0}^{M-1} d_\ell^{(K)} x^{2M-2\ell-2}.$$

This polynomial is independent of K in the stated range; the sum is empty when $M = 0$. Define the finite computational expression

$$\begin{aligned} Q_{M,N,K} &= \sum_{n=0}^{N-1} \{ (n+1/2)^{2M} T(n+1/2)^2 - P_M(n+1/2) \} \\ &\quad + \sum_{\ell=M}^{2K} d_\ell^{(K)} \zeta(2\ell+2-2M, a). \end{aligned} \tag{A.2}$$

Every Hurwitz zeta argument in the last sum is at least 2. The ordinarily convergent moment V_M satisfies

$$|V_M - Q_{M,N,K}| \leq C_K(1 + C_A)\zeta(2K+4-2M, a). \tag{A.3}$$

To verify this, for $x \geq a$ use $|A_K(x)| \leq C_A/x$ and (A.1) to obtain

$$x^{2M}|T(x)^2 - A_K(x)^2| \leq C_K(1 + C_A)x^{2M-2K-4}.$$

Summing this inequality over $x = n + 1/2$, $n \geq N$, proves (A.3). Expanding the finite square A_K^2 and subtracting P_M gives exactly the Hurwitz zeta tail in (A.2).

The accompanying script uses $N = 64$, $K = 30$, and 105 decimal working digits. Its recorded discrepancies from the exact Bernoulli formula and the analytic truncation envelopes are as follows:

M	Observed absolute discrepancy	Analytic truncation envelope
0	$9.911754463 \cdot 10^{-80}$	$1.015061650 \cdot 10^{-79}$
1	$4.203540245 \cdot 10^{-76}$	$4.304746532 \cdot 10^{-76}$
2	$1.784489779 \cdot 10^{-72}$	$1.827411085 \cdot 10^{-72}$
3	$7.583667641 \cdot 10^{-69}$	$7.765877807 \cdot 10^{-69}$
4	$3.226610534 \cdot 10^{-65}$	$3.304045785 \cdot 10^{-65}$
5	$1.374533946 \cdot 10^{-61}$	$1.407480267 \cdot 10^{-61}$
6	$5.863410357 \cdot 10^{-58}$	$6.003761793 \cdot 10^{-58}$

The script also verifies exactly the first centered Bernoulli pole test and its stated near-miss example, the formal translation recurrence through degree 14, the affine derivative relations for orders 2 through 8, nine Faulhaber block constants, and the exact moment coefficients through $M = 10$. These finite checks audit the formulas; they do not substitute for the all-order proofs.

References

- [1] ProveIt incoming report, *Exact Resonant Identities Beyond the Diagonal*, 11 October 2026. [Pinned source archive](#). Especially Sections 2–4 and 6–7, Questions 8 and 11.
- [2] ProveIt incoming report, *Independent Orders and Exact Reductions in Polylogarithm–Zeta Calculus*, 11 October 2026. [Pinned source archive](#). Independent Tornheim orders, boundary jets, and Section 7.
- [3] ProveIt incoming report, *Cubic Tornheim Derivatives and Harmonic Stieltjes Constants*, 11 October 2026. [Pinned source archive](#). Especially the cubic harmonic bridge and Section 7, Question 4.
- [4] ProveIt incoming report, *Periodic Stieltjes Collisions: Exact Contact Terms, Finite-Part Fubini Laws, and Gamma–Polylogarithm Primitives*, 10 October 2026. [Pinned source archive](#).
- [5] ProveIt incoming report, *Periodic Stieltjes Contact Calculus: All-Index Collision Corrections, Harmonic Derivative Anomalies, and Exact Zeta-Jet Integrals*, 10 October 2026. [Pinned source archive](#).
- [6] ProveIt incoming report, *Mixed Spectral Identities: Harmonic Newton Series, Dougall Jets, Unequal Twists, and Stieltjes Collisions*, 10 October 2026. [Pinned source archive](#). All-factor spectral completion and the three-factor contact specialization; internal paths are recorded in `SOURCE_AUDIT.md`.
- [7] O. R. Espinosa and V. H. Moll, *On some definite integrals involving the Hurwitz zeta function*, Ramanujan Journal **6** (2002), 159–188. <https://arxiv.org/abs/math/0012078>. The cited section numbers refer to the 34-page arXiv version.
- [8] L. Kargin, A. Dil, M. Cenkci, and M. Can, *On the Stieltjes constants with respect to harmonic zeta functions*, 2023. <https://arxiv.org/abs/2304.03517>. Related journal DOI: <https://doi.org/10.1016/j.jmaa.2023.127302>.
- [9] M. Can, L. Kargin, M. Cenkci, and A. Dil, *On certain harmonic zeta functions*, 2024. <https://arxiv.org/abs/2403.07123>. Version 1, especially its introductory harmonic-zeta Laurent normalizations.
- [10] J. M. Borwein and K. Dilcher, *Derivatives and fast evaluation of the Witten zeta function*, Ramanujan Journal **45** (2018), 413–432. <https://doi.org/10.1007/s11139-017-9890-9>. Author manuscript.
- [11] C. Hardouin and M. F. Singer, *Differential Galois theory of linear difference equations*, Mathematische Annalen **342** (2008), 333–377. <https://arxiv.org/abs/0801.1493>. Context for the additive-difference mechanism; the lemma used here is proved in full in Section 3.
- [12] NIST Digital Library of Mathematical Functions, §25.11, *Hurwitz Zeta Function*. <https://dlmf.nist.gov/25.11>. Fourier expansion, shift and argument-derivative identities, and Lerch’s formula.
- [13] NIST Digital Library of Mathematical Functions, §5.12, *Beta Function*. <https://dlmf.nist.gov/5.12>.
- [14] NIST Digital Library of Mathematical Functions, §5.11, *Asymptotic Expansions*. <https://dlmf.nist.gov/5.11>.
- [15] NIST Digital Library of Mathematical Functions, §25.12, *Polylogarithms*. <https://dlmf.nist.gov/25.12>. In particular Eq. 25.12.12.
- [16] NIST Digital Library of Mathematical Functions, §4.36, *Infinite Products and Partial Fractions*. <https://dlmf.nist.gov/4.36>. In particular the hyperbolic-cosecant expansion, Eq. 4.36.5.